

Limited-Memory Renewal Contracts: Budgeted Compression and Certified Joint Design

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Abstract. A renewal provider must select a limited execution alphabet and allocate an accepted obligation under participation caps, realization ceilings, and level-opening charges. We derive an exact target-crossing dynamic program that computes the entire memory–charge frontier conditional on branch targets. At saturation it solves joint catalog design in polynomial time with variable branch count and symbol budget. At an interior promise, a feasible weighted target net gives any prescribed additive accuracy at the original symbol budget when the branch count is fixed; its accuracy exponent is explicit. Local distortion certificates, charge-aware compression of price supports, and an inexact-oracle repair theorem provide complementary feasible policies without assuming zero duality gap. Exact continuous quadratic design and saturated Monge results are retained. Independent support enumeration and a separate uneliminated mixed-integer formulation validate the computation. Across thirty-six synthetic comparisons, the core target portfolio attains thirty-five original-budget optima; a retained charged counterexample motivates an explicit greedy safeguard. Heterogeneous scaling and adaptive-catalog experiments distinguish conditional performance from global guarantees. The resulting theory addresses limited-memory design without making additional symbols a prerequisite for approximation.

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1 Introduction

An execution interface may support only a few certified service levels even when a planner can distinguish many customer histories. Designing such an interface is not just a forecasting problem. The planner must select the levels, allocate an accepted aggregate obligation across branches, and respect both participation limits and the largest permitted realization. Enabling another level may also require certification or maintenance expenditure. We study these decisions jointly for a renewal provider with concave terminal reward and heterogeneous convex intermediate-service costs.

The central question is how to optimize within the stated symbol budget. An inexpensive pricing problem does not by itself answer this question: its supporting policies can require different alphabets, and the price bound can have a positive gap. Taking their union implements a useful resource-augmentation policy, but may exceed the permitted memory and incur unnecessary opening charges. We instead use the ordered interface twice. First, once the branch targets are specified, an eligibility-aware path dynamic program optimizes the alphabet at every symbol budget, including all opening charges. Second, a feasible net of branch targets converts this conditional optimizer into an additive approximation scheme for the original joint problem when the number of branches is fixed. Neither step randomizes the installed alphabet or relaxes the accepted promise.

The first result is an exact budgeted compression theorem. Consecutive installed levels partition the branches by their targets. An edge contributes either an eligible adjacent lottery or a fixed intermediate-service tail; the distinction depends on each branch’s realization ceiling. One dynamic program computes the entire conditional memory–value frontier. At a saturated aggregate obligation the targets are forced, so the same algorithm is a polynomial global catalog optimizer with arbitrary nonnegative level charges and heterogeneous realization ceilings. Away from saturation, conditional optimality is not joint optimality. The second result addresses this remaining target choice directly: a weighted target net with exact feasibility repair gives any prescribed additive accuracy without increasing the symbol budget. Its accuracy dependence has an exponent determined by the branch count, which we state explicitly rather than describing it as a polynomial algorithm for an unrestricted number of branches.

A complementary set of certificates makes the computation informative before such a net becomes affordable. Capped ideal targets give a Jensen upper bound. Optimizing their alphabet yields a same-budget lower policy and an instance-dependent distortion-and-charge certificate. Local interpolation errors replace a uniform mesh constant when the relevant targets and cells are known. In the absence of charges, uniform comparison books give first-order accuracy in the inverse symbol count, and second-order accuracy when the realization headroom accommodates adjacent lotteries. These bounds compare policies at the original budget; they do not require the additional symbols used by union recovery.

The price oracle remains valuable. We compress its feasible target mixtures at every desired budget, optimize the charge–distortion tradeoff rather than pay the whole union charge, and compare the result with existing feasible policies. An inexact-oracle theorem separates support suboptimality, price error, compression loss, and target-repair error. It requires certified price bounds and feasible endpoint policies, not an unjustified monotonicity rule for approximate optimizers. The exact continuous quadratic reduction, the saturated Monge results, and the institutional comparisons are retained as complementary parts of the

theory.

Our computational study distinguishes correctness, optimization quality, and application evidence. Exact conditional comparisons use an independent enumeration of supports and books. A second formulation keeps the original probability, intermediate-service, activation, and realization constraints in a mixed-integer model, without using the canonical response or either dynamic program. Tangent and secant envelopes provide numerical brackets that are compared with exact catalog optima. Across six seeds and three cap families, the core target portfolio attains the original-budget optimum in thirty-five of thirty-six instances; an explicitly retained charged counterexample is solved by the greedy fallback. Larger heterogeneous instances test conditional scaling, not unproved joint optimality. Adaptive-catalog comparisons use exact continuous references only in regimes where those references are available. All settings are synthetic; no field calibration or operational acceptance of additional memory is asserted.

1.1 Relation to established optimization methods

Finite policy graphs separate a rich history from a finite implementation interface (Zhang, 2012), while quantized feedback control studies the consequences of finite command sets (Fu, 2024). Optimal scalar quantization and dual quantization provide established methods for choosing levels and unbiased adjacent randomizations (Wu, 1991; Pagès and Wilbertz, 2012; Jourdain and Pagès, 2021). Our distortion is not nearest-neighbor error: a skipped eligible level changes an adjacent lottery, while an ineligible upper endpoint creates a pre-draw intermediate-service obligation. Its economic cost and feasible support therefore depend on the branch target and realization limit.

Interval dynamic programming is classical, including the scalar segmentation framework of Nielsen and Nock (2014). We do not claim the generic path recurrence, a target net, or convexification as new mechanisms. The structural content is the exact target-crossing decomposition under contractual timing and heterogeneous eligibility, and its use as a polynomial conditional oracle inside a same-budget global approximation scheme. Lagrangian bounds and constrained-path recovery are also established methods (Handler and Zang, 1980; Lemaréchal, 2001). Here their role is explicitly separate from feasibility: the union lemma makes mixtures implementable, whereas the new compression and target-net results retain the original resource cap. The quadratic stationary-face method supplies an exact, exponential reference for unrestricted continuous design rather than a competing polynomial-time claim (Del Pia et al., 2017).

The article develops the model, the conditional frontier, and the same-budget guarantees before reporting comparative computation. The appendices and electronic companion retain the continuous and saturated theory, detailed proofs, and reproducibility specifications. Earlier derivations and executed studies remain available in immutable source directories and in the complete predecessor reader package; the preservation map distinguishes archival evidence from newly executed experiments.

2 Model and Canonical Allocation

A public branch j is observed with probability $\pi_j > 0$, with $\sum_j \pi_j = 1$. It has a continuation cap $b_j \in (0, 1)$ and a convex nondecreasing intermediate-service cost h_j , with $h_j(0) = 0$. The common terminal reward

f is strictly increasing and concave on $[0, 1]$. Caps may repeat; write $\bar{b} = \sum_j \pi_j b_j$. An encoder selects a terminal command from one installed alphabet, and only its symbol reaches the common terminal gateway. The gateway cannot recover the observed branch, an inherited intermediate tier, or an uncharged shared random seed. Thus m distinct commands require $\lceil \log_2 m \rceil$ writable bits; read-only program constants and public-state storage are separate resources. The full institutional mechanism and saturated benchmarks are retained in the companion.

The provider selects a pre-draw tier y_j and a lottery C_j on the installed book, with operating payoff $\mathbb{E}f(C_j) - h_j(y_j)$. The accepted root obligation is $B \in [0, \bar{b}]$. Expected participation restricts the mean obligation, while a realization ceiling restricts every positive-probability draw. With a common risk tolerance $\delta \geq 0$,

$$t_j = y_j + \mathbb{E}C_j \leq b_j, \quad y_j + C_j \leq b_j + \delta \text{ almost surely}, \quad \sum_j \pi_j t_j = B. \quad (1)$$

A book c has at most m distinct levels in $[0, 1]$ and incurs $\sum_{z \in c} \rho(z)$, once per selected level. Let $\mathcal{V}_m^\delta(B; \rho)$ denote maximum operating payoff less this charge. A finite catalog restriction is denoted $\mathcal{V}_{m, \mathcal{A}}^\delta(B; \rho)$. The expectation cap is not replaced by the weaker realization ceiling. Taking $\delta \geq 1$ makes the realization restriction redundant at the canonical optimum below. Taking $\delta = 0$ gives the full pathwise value, even though that institution initially permits draw-specific intermediate service. Thus these are endpoints of one design problem, not three unrelated extensions.

2.1 Eligibility prefixes replace saturated targets

For a strictly ordered book $c = (c_1, \dots, c_s)$ and branch j , define

$$c^{j, \delta} = \{z \in c : z \leq b_j + \delta\}, \quad d_j = \max c^{j, \delta},$$

and let $F_{j, c}^\delta$ be the piecewise-affine interpolant of f on this prefix, including the singleton interpretation. The prefix depends on the cap and risk ceiling, not on the endogenous target.

Theorem 2.1 (Canonical allocation under a realization ceiling). *Suppose f is strictly increasing and concave and h_j is convex, nondecreasing, and zero at zero. A fixed book is feasible exactly when $c_1 \leq \min_j b_j$ and $c_1 \leq B \leq \bar{b}$. Its optimal branch response at target $c_1 \leq t \leq b_j$ is*

$$v_{j, c}^\delta(t) = F_{j, c}^\delta(\min\{t, d_j\}) - h_j((t - d_j)_+). \quad (2)$$

It is attained by $y = (t - d_j)_+$ and an adjacent lottery of mean $\min(t, d_j)$ in the eligible prefix. Consequently

$$\mathcal{V}_m^\delta(B; \rho) = \max_{\substack{|c| \leq m, \\ c_1 \leq t_j \leq b_j \\ \sum_j \pi_j t_j = B}} \left\{ \sum_j \pi_j v_{j, c}^\delta(t_j) - \sum_{z \in c} \rho(z) \right\}. \quad (3)$$

The fixed-book inner problem is concave. With zero charges, $\mathcal{V}_m^0(B; 0) = V_m^P(B)$ and $\mathcal{V}_m^\delta(B; 0) = V_m^E(B)$ for $\delta \geq 1$, at every feasible B .

Proof. Every used terminal level satisfies $C \leq b_j + \delta$ because $y \geq 0$. Writing $\mu = \mathbb{E}C$, the payoff is at most $F_{j,c}^\delta(\mu) - h_j(t - \mu)$. Both terms are nondecreasing as μ increases on $[c_1, \min(t, d_j)]$, so this upper bound is maximized at its right endpoint. If $t \leq d_j$, the adjacent lottery has $y = 0$ and all its support lies below $b_j + \delta$. If $t > d_j$, it is the singleton d_j and $y + d_j = t \leq b_j$. Thus the upper bound is attained without relaxing either constraint. The response is concave: its nonnegative interpolation slopes decrease, and its tail has the nonpositive slopes of $-h_j(t - d_j)$. The feasibility conditions follow by using the lowest level and allocating intermediate service; weighted targets fill the entire interval $[c_1, \bar{b}]$.

For pathwise participation with possibly draw-specific Y , all used C satisfy $C \leq b_j$. At fixed $t = \mathbb{E}(Y + C)$, concavity of f on the discrete support and Jensen's inequality for h_j give the same upper bound with $\delta = 0$ and $\mathbb{E}Y = t - \mu$. The canonical policy attains it and is pathwise feasible. Fresh decoder randomization can first be replaced by its conditional mean for each symbol: this increases terminal reward and preserves all the ceilings under the stipulated absence of an uncharged shared seed. Under expected participation the same mean argument has no eligibility truncation; $\delta \geq 1$ includes every level in $[0, 1]$. These observations establish the two endpoint identities and both directions of (3). $\square$

For quadratic primitives (31), chord $[u, v]$ has slope $r - q(u + v)/2$. On branch j its target capacity is

$$\left[\min\{b_j, v\} - u \right]_+ \mathbf{1}\{v \leq b_j + \delta\}.$$

Starting at $t_j = c_1$, fill these positive slopes in their common decreasing order, followed by zero-cost intermediate capacity and the quadratic tails. A water-level sweep over events $\gamma_j(b_j - d_j)_+$ fills the remaining targets. It uses $O(ks + k \log(k + 1))$ rational arithmetic operations, including the certificate

$$\sum_j \pi_j v_{j,c}^\delta(t_j) = \lambda B + \sum_j \pi_j \max_{c_1 \leq t \leq b_j} \{v_{j,c}^\delta(t) - \lambda t\}. \quad (4)$$

The right-hand branch maxima are independently checked at eligible knots, the cap, and the clipped tail stationary point. This multiplier certifies the allocation for that book; it is not by itself a global codebook certificate. The outer problem is solved next.

All canonical arguments continue to hold for branch-specific ceilings $\tau_j \geq b_j$ in place of $b_j + \delta$: the eligibility prefix is then $c \cap [0, \tau_j]$. We write $\mathcal{V}_{m,\mathcal{A}}^\tau(B; \rho)$ for this generalized catalog value. The new budgeted algorithms and numerical formulation use these heterogeneous ceilings. The retained continuous and institutional results use their stated common- δ specialization.

3 Exact Budgeted Compression at Fixed Targets

We first solve alphabet selection without changing the promised branch targets. This is both an independently meaningful design problem and an oracle for joint design. Let $\mathcal{A} = (a_1 < \dots < a_N)$, with charges $\rho_i \geq 0$, and fix $t_j \leq b_j$, $\sum_j \pi_j t_j = B$, and $a_1 \leq \min_j t_j$. Write $\tau_j \geq b_j$ for the realization ceiling; the renewal specialization is $\tau_j = b_j + \delta$. In this section $v_{j,c}$ denotes (2) with this branch-specific ceiling, and $R(c) = \sum_{a_i \in c} \rho_i$.

For two candidate levels $u < v$, define the target-crossing edge and stopping reward

$$A_t(u, v) = \sum_{j: u \leq t_j < v} \pi_j \begin{cases} f(u) + \frac{t_j - u}{v - u} \{f(v) - f(u)\}, & v \leq \tau_j, \\ f(u) - h_j(t_j - u), & v > \tau_j, \end{cases} \quad (5)$$

$$Z_t(u) = \sum_{j: t_j \geq u} \pi_j \{f(u) - h_j(t_j - u)\}. \quad (6)$$

The eligibility test is on v , not on its mean with u . If it fails, a branch uses the lower level and a fixed tier $t_j - u$; averaging an ineligible realization into a feasible mean would not implement the contract. Unlike the priced recurrence, (5) has no price-admissibility restriction on an arc.

Theorem 3.1 (Conditional memory–charge frontier). *For any feasible target vector t , define*

$$F_\ell(t) = \max_{\substack{c \subseteq \mathcal{A}, |c|=\ell \\ c_1 \leq \min_j t_j}} \left\{ \sum_j \pi_j v_{j,c}(t_j) - R(c) \right\}, \quad P_s(t) = \max_{1 \leq \ell \leq s} F_\ell(t).$$

Infeasible exact cardinalities have value $-\infty$. Set $H_1(i) = Z_t(a_i)$ and

$$H_\ell(i) = \max_{v > i} \{A_t(a_i, a_v) - \rho_v + H_{\ell-1}(v)\}. \quad (7)$$

Then $F_\ell(t) = \max_{a_i \leq \min_j t_j} \{H_\ell(i) - \rho_i\}$. All frontiers through budget $M \leq N$ and their feasible policies are computable in $O((k + M)N^2)$ arithmetic operations and $O(N^2 + MN)$ storage, given constant-time evaluations of f and h_j . Rational quadratic data give polynomial bit complexity. Every policy uses one fixed book, preserves the supplied targets exactly, commits intermediate service before its lottery, and respects every realization ceiling.

Proof. Take a feasible ordered book. Every target belongs to exactly one interval $[u, v)$ between consecutive levels, or to the final interval $[u, \infty)$. In the former case $u \leq t_j \leq b_j \leq \tau_j$, so the lower endpoint is eligible. If v is eligible, the canonical response is its chord at t_j . If v is ineligible, every later level is ineligible, and the highest eligible book level is u ; the response is the second line of (5). A final-interval branch has the response in (6). These intervals partition the branches, including targets equal to a selected level. The book's operating value is therefore the sum of its edge and stopping rewards.

The recurrence enumerates ordered paths with exactly ℓ vertices. It pays every charge once, with the first charge outside the recurrence. Conversely, each such path starting no higher than the smallest target defines a feasible book, and the preceding partition computes its exact value. Maximization proves both directions of the claimed equality. Taking the best exact cardinality gives an at-most budget and does not assume that an unused additional symbol should be purchased. There are $O(N^2)$ edges, each computed by scanning the branches, and $O(MN^2)$ transitions. Each rational path value is a sum of polynomially many input-derived rational terms, so ordinary rational arithmetic and comparison have polynomial bit complexity. $\square$

Corollary 3.2 (Global catalog design at saturation). *If $B = \bar{b}$, then $P_s(b) = \mathcal{V}_{s,\mathcal{A}}^\tau(\bar{b}; \rho)$ for every s . Hence joint catalog design at a saturated promise, including arbitrary nonnegative opening charges and*

branch-specific realization ceilings, has the complexity in Theorem 3.1, with both k and s variable.

Proof. Positive weights and $t_j \leq b_j$ imply $\sum_j \pi_j t_j = \bar{b}$ only when every $t_j = b_j$. There is no remaining target optimization. Apply the theorem. $\square$

This corollary is a same-budget global result, not a priced relaxation. At an interior promise, the theorem optimizes the entire alphabet conditional on t ; it does not certify that the selected target vector is jointly optimal. Section 4 resolves that target choice at fixed branch count. The faster saturated Monge construction in Appendix C concerns its stronger structural assumptions and is retained rather than replaced.

3.1 Compression, opening costs, and symbol prices

Take any candidate pool U , including a union of supported books. Applying the theorem with $\mathcal{A} = U$ gives the optimal sub-book at every allowed cardinality, not just a sequence of greedy deletions. Applying it on the whole catalog can only improve the conditional value. No nestedness of optimal books is required. In particular, comparing budgets $m, m+1, \dots, |U|$ gives the complete conditional memory–value frontier and identifies the value purchased by each additional symbol.

If an additional uniform certification fee $\kappa \geq 0$ is imposed, the exact conditional value is

$$\max_{1 \leq \ell \leq M} \{F_\ell(t) - \kappa \ell\}. \quad (8)$$

The upper envelope has finitely many breakpoints, obtained from the computed exact-cardinality values. Chosen cardinalities may be tied, but a largest optimal cardinality is nonincreasing in κ . Indeed, optimality at $\kappa_1 < \kappa_2$ and cardinalities ℓ_1, ℓ_2 gives $(\kappa_2 - \kappa_1)(\ell_1 - \ell_2) \geq 0$. This statement concerns cardinality, not nesting of the selected levels. General nonuniform charges are already included in the recurrence and do not require the uniform-fee interpretation.

The structural advantage over a raw union is that charges enter the optimization itself. A costly level may be discarded even when capacity remains. Nevertheless, an optimal conditional compression can be inferior to a policy with different branch targets. Our returned operational policy therefore compares valid candidates at the *same* budget and keeps the best, including a supplied incumbent. This safeguard never converts a heuristic target portfolio into a global optimality theorem.

4 Joint Approximation Without Additional Memory

The exact conditional oracle permits a global approximation of the original catalog problem without enumerating books or increasing their cardinality. The price is dependence on the dimension of the target vector. Let f be L_f -Lipschitz and each h_j be $L_{h,j}$ -Lipschitz on $[0, b_j]$, and set $L = \max\{L_f, \max_j L_{h,j}\}$. For the quadratic implementation, $L = \max\{r, \max_j \gamma_j\}$ is valid. These assumptions bound value changes, not the sensitivity of a selected alphabet.

Lemma 4.1 (Target sensitivity and exact repair). *For every feasible book c , its branch response is L -Lipschitz on $[c_1, b_j]$. Given $u_j \in [c_1, b_j]$ and a feasible root promise B , bounded transfers produce $t_j \in [c_1, b_j]$ with*

$\sum_j \pi_j t_j = B$ and

$$\sum_j \pi_j |t_j - u_j| = \left| B - \sum_j \pi_j u_j \right|. \quad (9)$$

The operating payoff decreases by at most L times the right-hand side. With rational inputs, both transfers and canonical lotteries are rational and satisfy the original constraints exactly.

Proof. Chord slopes of f have absolute value at most L_f ; the final shortfall tail has slopes bounded by $L_{h,j}$. Their continuous concatenation is therefore L -Lipschitz. If the residual is positive, increase targets within their remaining capacities; if it is negative, decrease them toward c_1 . Total available capacity is sufficient because $B \in [c_1, \bar{b}]$. Stop when the residual is zero. All changes have the same sign, proving (9); the payoff bound follows by the weighted Lipschitz inequality. Reconstruct the lotteries after repairing targets, rather than rounding their probabilities separately. The canonical rule then enforces $y_j + C_j \leq \tau_j$ for each realization. $\square$

Theorem 4.2 (Same-budget additive scheme at fixed branch count). *For a feasible finite catalog and any $\varepsilon > 0$, a target-net algorithm returns one feasible book of at most m levels and net value J satisfying*

$$\mathcal{V}_{m,\mathcal{A}}^\tau(B; \rho) - \varepsilon \leq J \leq \mathcal{V}_{m,\mathcal{A}}^\tau(B; \rho). \quad (10)$$

For $k \geq 2$, set $\eta = \varepsilon / [2L(k-1)]$. At most

$$N(1 + 1/\eta)^{k-1} \quad (11)$$

target profiles need be optimized. Thus a safe arithmetic bound is $O((k+m)N^3(1+2L(k-1)/\varepsilon)^{k-1})$. For $k = 1$, apply the conditional oracle once at $t_1 = B$. The guarantee allows arbitrary nonnegative catalog charges and arbitrary ceilings $\tau_j \geq b_j$; neither is approximated. For rational quadratic data the algorithm is polynomial in input bit length and the displayed numerical accuracy parameter when k is fixed. It is not an FPTAS with variable k and does not claim polynomial dependence on the binary length of $1/\varepsilon$.

Proof. Enumerate a possible lowest level $a \in \mathcal{A}$ with $a \leq a_* := \min\{B, \min_j b_j\}$. Restrict the conditional candidate pool to levels at least a . In weighted target coordinates, let $x_j = \pi_j(t_j - a)$, $d_j = \pi_j(b_j - a)$, and $C = B - a$. Choose one distinguished branch, numbered k only for this proof. Enumerate multiples of η in $[0, d_j]$ for $j < k$, discarding tuples whose sum exceeds C . Set

$$x_k = \min\{d_k, C - \sum_{j < k} x_j\}.$$

Any remaining deficit is nonnegative; fill it by bounded increases to the coordinates $j < k$. Their total available capacity is sufficient because $C \leq \sum_j d_j$. This constructs a feasible profile $t_j = a + x_j/\pi_j$. Run Theorem 3.1 at budget m and retain the best policy over all anchors and profiles.

Let (c^*, t^*) be a global catalog optimum, which exists by compactness of each fixed-book allocation problem and finiteness of the catalog. Its lowest level $a = c_1^*$ is enumerated. For $j < k$, the enumeration includes $x_j = \eta \lfloor x_j^*/\eta \rfloor$. Their aggregate decrease is $D < (k-1)\eta$. The constructed last coordinate is at least

x_k^* and increases by at most D . If it reaches its cap, the subsequent increases on the other coordinates use only the remaining deficit. Consequently, the final weighted target distance from t^* is at most $2D \leq 2(k-1)\eta$. All final targets remain at least a , so the *same* book c^* is feasible. Lemma 4.1 bounds its payoff loss by ε ; its opening charge is unchanged. The conditional optimizer does no worse. Maximizing over profiles proves (10).

Each of the $k-1$ coordinates has at most $1 + d_j/\eta \leq 1 + 1/\eta$ possibilities. There are at most N anchors. Applying the conditional-oracle complexity gives the stated bound. The construction uses rational flooring, addition, comparison, and the rational oracle, so the bit-complexity conclusion follows under the stated representation assumptions. $\square$

The scheme is useful when the number of distinct contract branches is small or repeated homogeneous branches have been aggregated. Its exponential accuracy dependence on k is not concealed by the polynomial dependence on the symbol budget. At a large branch count we use selected target profiles and certified upper bounds rather than declare the net completed. An interrupted enumeration returns a feasible incumbent, but the implementation deliberately withholds the requested ε guarantee until coverage is complete.

This result addresses the original-budget difficulty differently from union recovery. It pays for accuracy in target-space computation, not extra installed levels. At saturation, Corollary 3.2 avoids target-space enumeration altogether. For a general interior promise, selected profiles can be polished by alternating exact inner allocation and exact conditional book design. This preserves feasibility and nondecreases value at every step, but a fixed iteration cap or stationary pair has no global-optimality implication. The comparative study records this distinction explicitly.

5 Local Distortion and Same-Budget Certificates

The target net provides a global guarantee when fully enumerated. A complementary certificate can be evaluated with just one conditional dynamic program. Let

$$t_j^0 = \min\{b_j, z\}, \quad \sum_j \pi_j t_j^0 = B, \quad W_0 = \sum_j \pi_j f(t_j^0).$$

At $B = \bar{b}$ take $t^0 = b$. The water level is otherwise found by sorting caps. Let $\rho_* = \min_{a_i \leq a_*} \rho_i$, where $a_* = \min\{B, \min_j b_j\}$. Feasibility guarantees that this set is nonempty.

Theorem 5.1 (Distortion–charge certificate). *For every symbol budget $s \geq 1$, define*

$$C_s = \min_{\substack{c \subseteq \mathcal{A}, |c| \leq s \\ c_1 \leq \min_j t_j^0}} \left[\sum_j \pi_j \{f(t_j^0) - v_{j,c}(t_j^0)\} + R(c) \right].$$

The conditional dynamic program computes $C_s = W_0 - P_s(t^0)$ exactly, and

$$P_s(t^0) \leq \mathcal{V}_{s,\mathcal{A}}^\tau(B; \rho) \leq W_0 - \rho_*. \quad (12)$$

Thus $C_s - \rho_*$ is an explicit original-budget optimality-gap certificate. Any other valid upper bound for that budget can replace the right-hand side by its minimum with $W_0 - \rho_*$. A larger-budget policy value is not such an upper bound.

Proof. For any policy, $\mathbb{E}C_j \leq t_j$ and Jensen's inequality and monotonicity give $\mathbb{E}f(C_j) - h_j(y_j) \leq f(t_j)$. Maximizing the sum of these ideal rewards under the root equality and caps gives t^0 : their marginal rewards are equal on uncapped branches, and weakly larger on capped branches. This remains an optimum when f is not strictly concave. Every feasible nonempty book contains a level at most a_* and pays at least ρ_* . Hence its net value is at most $W_0 - \rho_*$. Conversely, the conditional optimal policy at t^0 is feasible at budget s . Finally, minimizing the displayed distortion plus charge is exactly the conditional objective subtracted from W_0 . $\square$

The certificate remains valid with nonuniform or large fees, because those fees are optimized rather than charged after an uncontrolled expansion. It can still be loose: C_s includes unavoidable finite-memory distortion, the cost of the selected levels, and any loss caused by fixing t^0 . We report its numerical size instead of asserting that it is uniformly small relative to the optimum. A multiplicative guarantee is particularly inappropriate when the optimal net payoff is zero or negative.

5.1 Local errors, adaptive candidates, and constructive rates

For a twice differentiable f , suppose $-f'' \leq M_{uv}$ on $[u, v]$. If an eligible pair brackets a target t , its interpolation error satisfies

$$0 \leq f(t) - \left[f(u) + \frac{t-u}{v-u}(f(v) - f(u)) \right] \leq \frac{M_{uv}}{2}(t-u)(v-t). \quad (13)$$

This follows by comparing f with its chord plus $M_{uv}(t-u)(v-t)/2$, whose difference has nonnegative second derivative and zero endpoint values. If the next selected level is ineligible or no upper level exists, the exact local deficit is

$$f(t) - f(u) + h_j(t-u), \quad (14)$$

not a second-order chord error. Summing these local quantities with the actual selected-level charges yields C_s for the chosen book. A piecewise curvature majorant can also be placed directly in the target-crossing recurrence to minimize a certified distortion bound. It must retain the first-order expression (14) on ineligible edges.

An adaptive candidate pool can include ideal target values and refine intervals that matter for (13)–(14). The resulting certificate is evaluated from the final pool and policy. We make no uniform convergence-rate claim for the greedy midpoint refinement used in the experiments. In particular, adding a level just above an eligibility threshold need not help the affected branch. Candidate selection and the exact optimization over those candidates are distinct tasks.

Proposition 5.2 (Original-budget comparison books). *Let $m \geq 2$, $h = 1/(m-1)$, and suppose the catalog*

contains the uniform m -point book $G = \{0, h, \dots, 1\}$. With zero charges, the conditional policy at t^0 satisfies

$$\mathcal{V}_{m,\mathcal{A}}^\tau(B; 0) - P_m(t^0) \leq L_f h + \sum_j \pi_j h_j(h), \quad (15)$$

$$\mathcal{V}_{m,\mathcal{A}}^\tau(B; 0) - P_m(t^0) \leq Mh^2/8 \quad \text{if } \min_j(\tau_j - b_j) \geq h \text{ and } -f'' \leq M. \quad (16)$$

In (15), $h_j(h)$ may be replaced by $h_j(\min\{h, b_j\})$ if costs are defined only on $[0, b_j]$. The same bounds hold relative to the uncharged continuous m -level optimum. With charges, adding $R(G) - \rho_*$ to the right-hand side gives the corresponding catalog comparison. If the distinct ideal targets are themselves available and number at most m , the uncharged policy attains W_0 and is globally optimal, for any realization headroom.

Proof. At each target, round its terminal level down to the nearest point of G and supply the difference before execution. The difference is at most h , the target and root promise are unchanged, and every realized total equals the target, hence is no larger than b_j . This proves (15) by comparison with W_0 . Under the headroom assumption the upper neighbor is no higher than $t_j^0 + h \leq b_j + h \leq \tau_j$, so the adjacent unbiased lottery is feasible with no intermediate service. Apply (13) and $(t - u)(v - t) \leq h^2/4$. The conditional optimum on the whole catalog is no worse than either comparison policy. Since W_0 also bounds continuous design, the continuous comparison follows. With fees, the comparison book pays $R(G)$ while the catalog upper bound subtracts ρ_* . Finally, installing the distinct ideal targets realizes them deterministically with no intermediate service, attaining the ideal upper bound. $\square$

These are explicit same-budget guarantees, not claims that uniform catalogs are always efficient. Their approximation parameter is the number of installed symbols, whereas Theorem 4.2 controls computational accuracy at a fixed symbol budget. The local certificate and the direct continuous comparisons below distinguish the two effects. In particular, driving a price tolerance to zero cannot remove finite-catalog distortion or the original-budget price gap.

6 Eligibility-Prefix Decomposition

The closed-cell formulation establishes exact continuous attainability. We now exploit the ordered interface to avoid enumerating either branch modes or catalog subsets in a price subproblem. The resulting bounds support finite global search and remain valid when that search stops early. This distinction matters: a price relaxation is not, without an additional equality, the original optimum.

6.1 A reusable ordered-interface allocation class

Let $\mathcal{A} = (a_1 < \dots < a_N) \subset [0, 1]$ be a catalog with nonnegative opening charges ρ_i . The common reward f is strictly increasing and concave. Branch j has weight π_j , target cap b_j , nondecreasing convex shortfall cost h_j with $h_j(0) = 0$, and an eligibility threshold $\tau_j \geq b_j$. Its response is (2) with eligible levels $c \cap [0, \tau_j]$. The renewal model has $\tau_j = b_j + \delta$; the theorem below permits distinct thresholds without requiring a common overrun tolerance. This abstract class needs no acceptance or renewal interpretation. It

describes any ordered shared interface with a common concave output reward, separable shortfall costs, and these target and support restrictions.

Write $a_* = \min\{B, \min_j b_j\}$. Assume $a_1 \leq a_*$, since otherwise no feasible book exists, and replace m by $\min\{m, N\}$. Restrict books to $c_1 \leq a_*$; this retains exactly those books that can meet the root promise. At a real price λ , define

$$g_\lambda(x) = f(x) - \lambda x, \quad \phi_j(z; \lambda) = \max_{0 \leq y \leq z} \{-h_j(y) - \lambda y\}, \quad (17)$$

$$D(\lambda) = \lambda B + \max_{\substack{\emptyset \neq c \subset \mathcal{A}, |c| \leq m \\ c_1 \leq a_*}} \left[\sum_j \pi_j \max_{c_1 \leq t \leq b_j} \{v_{j,c}(t) - \lambda t\} - \sum_{a_i \in c} \rho_i \right]. \quad (18)$$

Weak duality gives $\mathcal{V}_{m, \mathcal{A}}^\delta(B; \rho) \leq D(\lambda)$ in the renewal specialization. For quadratic shortfall costs, (17) is evaluated at $y = 0$ if $\lambda \geq 0$, and at $y = \min\{z, -\lambda/\gamma_j\}$ otherwise; when $\gamma_j = 0$ and $\lambda < 0$, use $y = z$.

Call an arc $u < v$ admissible when $g_\lambda(v) \geq g_\lambda(u)$. Define its reward and the reward for stopping at u by

$$E_\lambda(u, v) = \sum_{j: u \leq b_j < v} \pi_j \begin{cases} g_\lambda(u) + \frac{b_j - u}{v - u} [g_\lambda(v) - g_\lambda(u)], & v \leq \tau_j, \\ g_\lambda(u) + \phi_j(b_j - u; \lambda), & v > \tau_j, \end{cases} \quad (19)$$

$$T_\lambda(u) = \sum_{j: b_j \geq u} \pi_j [g_\lambda(u) + \phi_j(b_j - u; \lambda)]. \quad (20)$$

An edge accounts only for branches whose target caps it crosses. It therefore carries a contractual allocation value, not an ordinary nearest-neighbor quantization error.

Theorem 6.1 (Polynomial price decomposition). *For every λ , some maximizer in (18) has nondecreasing g_λ along its ordered book. Its priced operating value is the sum of (19) over consecutive levels plus (20) at the last level. Set*

$$W_1(i) = T_\lambda(a_i), \quad (21)$$

$$W_r(i) = \max \left\{ T_\lambda(a_i), \max_{\substack{\ell > i \\ g_\lambda(a_\ell) \geq g_\lambda(a_i)}} [E_\lambda(a_i, a_\ell) - \rho_\ell + W_{r-1}(\ell)] \right\}.$$

Then $D(\lambda) = \lambda B + \max_{a_i \leq a_*} [W_m(i) - \rho_i]$. With constant-time evaluation of ϕ_j , computing a maximizing book and this bound takes $O((k+m)N^2)$ arithmetic operations and $O(N^2 + mN)$ storage, allowing both k and m to vary. Rational quadratic data and a rational price give an exact rational algorithm with polynomial bit complexity. General costs use at most $O(kN^2)$ calls to the scalar oracle (17) instead.

Proof. If $\lambda < 0$, strict increase of f makes g_λ strictly increasing. If $\lambda \geq 0$, intermediate service cannot improve priced payoff, and a branch maximizes the interpolant of g_λ on its eligible prefix, truncated at its target cap. Concavity implies that the values along any book increase up to their first maximum and never subsequently increase. All levels after that first maximum may be deleted: for a branch below the maximizing level, either that level is ineligible and all later levels are also ineligible, or later chords cannot improve the

concave interpolant before the cap; for a branch above it, no later value beats the maximum. Nonnegative charges make deletion weakly beneficial. The retained book has only admissible arcs.

Consider $b_j \in [u, v)$ on such a book. If $v \leq \tau_j$, the best priced target is b_j on the chord $[u, v]$. For $\lambda \geq 0$ its slope is nonnegative and a singleton tail contributes no positive adjustment. For $\lambda < 0$, a singleton tail contributes at most $-\lambda(b_j - u)$, whereas the chord contributes this amount plus a strictly positive reward increment when $b_j > u$. If $v > \tau_j$, every later level is ineligible; the optimum is a singleton at u followed by the best shortfall in (17). Branches beyond the final level have the same singleton-tail form. These disjoint cap intervals cover all branches because the first level is at most $\min_j b_j$. This proves the decomposition, including equality cases.

The acyclic recurrence chooses either to stop or to append the next admissible level, paying each selected charge once. It enumerates paths implicitly rather than enumerating books. Computing edges costs $O(kN^2)$; the dynamic-programming layers cost $O(mN^2)$. For rational quadratic data, every scalar oracle value uses a bounded number of rational operations, and each path value sums at most $O(km)$ such terms. Numerator and denominator lengths are polynomial in the input bit length; comparisons and path reconstruction therefore preserve polynomial bit complexity. $\square$

Pricing and constrained shortest-path bounds are established optimization mechanisms (Handler and Zang, 1980). The model-specific result here is that one price makes an otherwise nonseparable shared-book allocation decompose into these cap-crossing edges, despite heterogeneous shortfall costs, branch-specific eligibility, and nonsaturated targets. The theorem does not assert a polynomial algorithm for the unrelaxed joint problem or exchange its maximization with minimization over prices.

6.2 Prefix bounds and checkable global search

A node is a mandatory initial segment $p = (i_1, \dots, i_s)$ of a book; its descendants may append only larger indices. If the priced values along p never decrease, its exact priced completion value is

$$U_p(\lambda) = \lambda B + \sum_{h=1}^{s-1} E_\lambda(a_{i_h}, a_{i_{h+1}}) - \sum_{h=1}^s \rho_{i_h} + W_{m-s+1}(i_s). \quad (22)$$

If a decrease occurs in p , concavity places its first priced maximum within p and every larger new level is useless to the priced allocation. In that case, use the fixed-prefix branch-support value, with every mandatory charge retained. This is again the exact priced maximum over completions; additional nonnegative charges cannot help. The empty prefix uses (18).

Theorem 6.2 (Finite search and an anytime global interval). *Fix any nonempty finite set of prices. At each prefix use the smallest corresponding $U_p(\lambda)$, evaluate feasible books by the exact inner allocator, and retain the largest feasible net value L . Partition an unpruned node into its own book, when nonempty, and all one-index extensions. Prune a node only when its upper bound is at most L . This procedure terminates after finitely many prefixes and returns the exact catalog optimum. At any interruption it returns*

$$L \leq \mathcal{V}_{m, \mathcal{A}}^\delta(B; \rho) \leq U = \max\{L, \max_{p \text{ unresolved}} \min_{\lambda} U_p(\lambda)\}. \quad (23)$$

For a uniform mesh of spacing h , Theorem D.1 further yields the continuous interval $[L, U + K_m h]$. The optimization gap $U - L$ and discretization error $K_m h$ are separate and observable.

Proof. The preceding completion argument proves each price bound, so their minimum is valid by weak duality. Every feasible book either equals a nonempty node or has a unique next index. Hence the prescribed children and the node's own book are a disjoint exhaustive partition; book size is at most m . A pruned subtree cannot improve the incumbent. The maximum of unresolved upper bounds and the incumbent therefore bounds every feasible book at every stage. If all subtrees have been resolved, the two bounds coincide. The mesh transport is a separate one-sided inequality between the continuous optimum and the exact catalog optimum, so substituting the catalog upper bound proves the last claim. $\square$

A global witness consists of the feasible winning lottery, every split and its complete child list, upper bounds and prices at pruned or unresolved nodes, and a fixed-book supporting price for each explicitly evaluated node book. A checker can rebuild price tables and verify the partition without replaying the search order or trusting the incumbent-selection code. This is a stronger object than an inner multiplier or a source hash. Equality of one root price bound with a feasible payoff already certifies global catalog optimality without any branching. Worst-case search is still exponential in the alphabet budget; neither this result nor the mesh transfer is an FPTAS claim.

Additional decomposition contribution. Theorems 6.1–6.2 use classical Lagrangian bounds and an acyclic path recurrence. Their structural increment is the exact eligibility-aware cap-crossing reward and prefix-completion dominance for nonsaturated allocation with heterogeneous shortfall costs. The generic search framework is not claimed as new. Proposition D.3 establishes sharpness for uniform catalogs, not a lower bound for adaptive grids. The continuous stationary-face mechanism remains generic; its contractual closed-cell reduction remains the structural contribution.

7 Charge-Aware Recovery from Inexact Price Supports

The polynomial price oracle in Section 6 remains a useful source of target profiles and upper bounds. We now state exactly what survives an inexact oracle, and how a fixed memory budget enters the recovery guarantee. For a feasible endpoint policy $i \in \{-, +\}$, let c^i , t^i , J_i , and $S_i = \sum_j \pi_j t_j^i$ be its book, targets, net payoff, and aggregate target. It need not satisfy the desired root promise B , but satisfies all branch constraints. Its book has at most m levels and its lowest level is at most a_* . At prices $\lambda_- \leq \lambda_+$ assume certified bounds $u_i \geq D(\lambda_i)$, and write

$$\zeta_i = u_i - \{J_i + \lambda_i(B - S_i)\} \geq 0. \quad (24)$$

This defect includes both optimization error and conservatism in the certified upper bound. An approximate feasible solution alone, without an upper bound, does not certify ζ_i .

Suppose $S_- > B > S_+$. Define

$$\theta = \frac{B - S_+}{S_- - S_+}, \quad t^U = \theta t^- + (1 - \theta)t^+, \quad c^U = c^- \cup c^+, \quad R_{\text{mix}} = \theta R(c^-) + (1 - \theta)R(c^+),$$

and

$$e = (\lambda_+ - \lambda_-) \frac{(S_- - B)(B - S_+)}{S_- - S_+}, \quad U = \min\{u_-, u_+\}. \quad (25)$$

Here U is the minimum of two valid endpoint bounds, not necessarily the minimized Lagrangian dual value. Endpoint equality can be handled by returning that endpoint policy; the strict bracket avoids division by zero.

Theorem 7.1 (Budgeted compression with oracle defects). *For any feasible sub-book $d \subseteq c^U$ with $|d| \leq s$ at the fixed mixed targets, put*

$$\Delta(d) = \sum_j \pi_j \{v_{j,c^U}(t_j^U) - v_{j,d}(t_j^U)\} \geq 0, \quad \Gamma_s = \min_d \{\Delta(d) + R(d) - R_{\text{mix}}\}.$$

Theorem 3.1 computes the minimizer and its policy. Its net payoff satisfies

$$J_s \geq U - e - \theta \zeta_- - (1 - \theta) \zeta_+ - \Gamma_s. \quad (26)$$

The policy has at most s symbols and preserves the mixed branch targets, their aggregate root promise, and every realization ceiling. The original budget is obtained by setting $s = m$. Optimizing over the full catalog instead of the union, or returning a better feasible incumbent at the same budget, preserves the bound. For rounded mixed targets, let r_{mix} bound their weighted distance from t^U after clipping to the selected book extquotesingles feasible target intervals, and let r_{eq} be their root residual. A further allowance $L(r_{\text{mix}} + r_{\text{eq}})$ suffices after exact repair.

Proof. The union weakly improves the branch response at each endpoint target because it contains both old books. Its response is concave in the target, so

$$\sum_j \pi_j v_{j,c^U}(t_j^U) \geq \theta [J_- + R(c^-)] + (1 - \theta) [J_+ + R(c^+)].$$

This compares operating values before subtracting installation charges. Direct expansion of (24) gives

$$\theta u_- + (1 - \theta) u_+ = \theta J_- + (1 - \theta) J_+ + e + \theta \zeta_- + (1 - \theta) \zeta_+.$$

The left side is at least U . Subtracting $\Delta(d) + R(d)$ from the union operating value proves (26) for any d , and minimizing that subtraction is precisely fixed-target book optimization. A feasible singleton always exists in the union: its smallest level is no higher than either endpoint's targets, hence no higher than each mixed target. Canonical reconstruction proves all feasibility claims. The repair loss follows by first bounding the target perturbation and then applying the previous lemma to the residual; installed-level charges do not change. $\square$

Unlike the union excess charge, Γ_s includes an *optimized* distortion–charge tradeoff at the actual returned budget. It may be positive, zero, or negative, and the bound can still be loose if compression is costly. We do not relabel it as a uniform relative approximation. The independently available net scheme gives the unconditional additive same-budget guarantee at fixed branch count; the compressed-support bound provides

a computable instance-specific certificate at large branch counts.

For exact price optimizers, $\zeta_i = 0$ when their price bounds are exact. Optimality at two prices gives the explicit tie-safe monotonicity argument

$$J_1 - \lambda_1 S_1 \geq J_2 - \lambda_1 S_2, \quad J_2 - \lambda_2 S_2 \geq J_1 - \lambda_2 S_1 \implies (\lambda_2 - \lambda_1)(S_1 - S_2) \geq 0.$$

Arbitrary selections among exact tied optima therefore preserve the ordering of supported totals. Approximate optimizers need not do so. The inexact theorem assumes and checks a feasible bracket rather than extending this monotonicity assertion incorrectly. It establishes a value bound conditional on certified endpoint data, not an oracle-call complexity bound for an unspecified approximate solver.

7.1 Relation to union augmentation and its strict gap

At $s = |c^U|$, choosing the whole union gives $\Delta = 0$ and $R(c^U) - R_{\text{mix}} = \Omega$; (26) recovers the charged union guarantee. For zero charges and exact supports, the usual price-bracket tolerance produces an at-most- $2m$ policy within the requested additive tolerance of the original-budget optimum. The full construction and continuous mesh transfer remain in the companion. They are bicriteria results, distinct from the original-budget net scheme and saturation theorem.

The exact two-branch example has caps $(1/4, 3/4)$, equal weights, curvatures $(1, 2)$, $f(x) = 2x - x^2/2$, catalog $\{i/8 : 0 \leq i \leq 8\}$, $m = 2$, $B = 2/5$, and $\delta = 0$. Its original-budget value is $45/64$, whereas the minimum price bound is $453/640$. The gap $3/640$ is $1/150$ of the original value. A three-level book $(1/4, 1/2, 5/8)$ attains the price value. This is a relaxation-gap example, not evidence that the factor two is necessary. The conditional frontier quantifies every compression budget for its mixed targets; the joint net scheme can instead approximate the two-level optimum. No factor-two necessity claim or same-budget strong-duality claim is used anywhere in these arguments.

8 Comparative Computation

The study separates exact conditional verification, joint optimization quality, independent model validation, and scaling. Every input, rational policy, seed, budget frontier, numerical solver result, and repeated timing is retained. The experiments are synthetic and make no population or field-performance claim. Tables 1–4 summarize executed files generated by the accompanying scripts, not manually entered performance claims.

8.1 Conditional correctness and the global target net

Six declared seeds produce 144 small instances with distinct and repeated caps, heterogeneous costs, several risk tolerances, and nonnegative charges. For each, an independent routine enumerates catalog subsets and all singleton and two-point supports at the supplied targets. It does not import the new recurrence, the old price decomposition, or the canonical reconstruction. All 288 tested budget values agree exactly with the conditional dynamic program. The suite also checks 144 target repairs, fifteen comparison-book rate

cases, four endpoint cases, ten semantic corruption rejections, and five invalid-input rejections. These are regression and theorem checks, not optimization-performance evidence.

The target-net scheme is tested separately on twenty two- and three-branch settings, including nonuniform charges. Each completed run is compared with an exact global catalog optimum obtained by enumerating books and solving their concave inner allocations. The actual loss is no larger than the requested additive tolerance in every run; the largest observed loss is 0.0498046875. Together these runs evaluate 385 feasible profiles, with at most 94 in a run. This modest experiment verifies the net construction and its coverage certificate; it does not disguise the exponential dependence on the number of branches. An interrupted run explicitly returns no global accuracy certificate.

8.2 Original and enlarged budget comparisons

The comparative sample has six seeds, three cap families (spread, clustered, and repeated), and both free and charged interfaces, for 36 instances. It varies weights, shortfall curvatures, risk tolerance, promise saturation, and arbitrary nonnegative charges. Each instance has three or four branches, seven or nine catalog levels, and original budget two or three. Exact catalog optima are computed at every budget through twice the original budget. This gives the original optimum, the optimum at the actual union size, and the full enlarged-budget reference rather than treating an original-budget price bound as an enlarged-budget optimality gap.

The core candidate method uses three feasible target profiles: capped ideal targets, the two-price mixed targets, and the original-budget incumbent’s targets. For each profile it computes the entire conditional frontier, then uses at most twelve monotone allocation–book improvement steps per budget. This finite target portfolio is a heuristic for joint design, not a substitute for the completed target net. Baselines are the old two-price recovery, greedy level addition with exact inner reallocation, greedy deletion of union levels with reallocation, and direct exact catalog optimization. The operational recommendation retains the better of the core method and the greedy policy at each budget. All component policies remain separately reported.

The core method attains the original-budget exact optimum on all eighteen free instances and seventeen of eighteen charged instances. It improves on greedy addition in ten cases; greedy addition is better in one. The charged exception, seed 205019 in the spread family, is important: the core target portfolio is 101/3600 below the optimum 1373/3600, while the greedy policy attains that optimum. The safeguard therefore attains all 36 original-budget optima in this sample. This observation is not a universal guarantee for the safeguard, nor is the exception removed from the data. It demonstrates why conditional global optimality must not be presented as global target optimality.

Two instances improve strictly on the old original-budget incumbent. Across all instances, endpoint books are nested in 23 cases; the largest observed union has four levels. The data retain actual union size, overlap, nonnegative expansion $\max\{0, |c^U| - m\}$, percentage expansion, each budget’s joint reference, and union loss at its own size. These outcomes do not establish that a factor-two expansion is necessary. The new conditional frontier instead answers how much each permitted budget can recover for the supplied targets, and the exact small references show the remaining target-selection loss.

8.3 A genuinely different optimization formulation

For twelve selected input instances, we solve the original uneliminated model at two budgets, giving 24 comparisons and 48 mixed-integer envelope solves. The model retains terminal probabilities, binary level activations, binary branch supports, a pre-draw intermediate tier, the exact root equality, and the implication that every positive-probability realization satisfies $y_j + a_i \leq \tau_j$. It contains neither target-crossing recurrence nor the canonical response formula. Section EC.13 gives the complete formulation.

Quadratic costs are bounded by tangent and secant piecewise-linear envelopes. The former gives a maximization upper problem; the latter gives a conservative feasible lower problem. Each uses 32 segments per branch, with an analytic envelope-error allowance. SciPy’s mixed-integer interface and HiGHS solve these models in floating-point arithmetic (Virtanen et al., 2020; Huangfu and Hall, 2018). In all 24 comparisons the exact catalog value lies between the numerical brackets within the declared 10^{-7} comparison tolerance, both solves terminate with reported optimal status, and direct model residuals are below 10^{-6} . The tangent-to-secant width is at most twice the envelope allowance plus 2×10^{-6} . These are independently formulated *numerical* checks, not exact rational proofs of global optimality. The exact comparisons and rational policy checks supply different evidence and are labeled accordingly.

8.4 Scaling, local certificates, and continuous references

Twelve further cases use two seeds, branch counts 32, 128, and 256, catalog sizes 17, 33, and 65, and budgets four or eight. Unlike the earlier deterministic family, caps, weights, curvatures, charges, and branch-specific realization headroom vary randomly. Each computes the ideal-target conditional frontier three times. Table 3 reports the median of these timings and the same-budget Jensen gap. These runs establish conditional scalability and feasible value certificates, not global optimization of a 256-dimensional target space. Timing replicates measure implementation variability on one machine and are not independent draws of an applied population. Stored dynamic-programming bit lengths exclude unrecorded arithmetic temporaries. The archived process peak memory is cumulative operating-system resident memory, not a sub-megabyte allocation-tracing statistic.

The continuous comparison uses four explicitly solvable regimes. In an interior two-branch case, caps $(3/10, 9/10)$, weights $(2/5, 3/5)$, zero headroom, and promise $66/125$ give ideal targets $(3/10, 17/25)$. A four-candidate adaptive pool containing these targets attains the continuous two-symbol optimum exactly. The uniform 65-point catalog still has a loss of 0.0029909375. For a saturated off-cap-top example, midpoint refinement reaches the exact continuous optimum with six candidates, whereas the nine-point uniform pool also reaches it. Two additional seven-branch saturated cases use the retained exact continuous optimizer as reference; target-adaptive pools with nine candidates attain their optima, while uniform pools require 33 points among the tested resolutions. All these comparisons keep the installed-symbol budget fixed. They illustrate the advantage of aligning candidate points with the problem, not a universal rate for adaptive refinement. Their exact-reference errors are distinct from the Jensen certificate, which can include intrinsic memory loss.

Finally, ten deliberately suboptimal support-policy pairs verify the inexact-price identity with positive

oracle defects, rather than merely retest exact optimizers. The target-repair tests reconstruct rational lotteries and check realized totals directly. Neither exercise validates unknown estimated model parameters. For uncertain caps or realization limits, a deployment must first supply conservative feasible limits; arbitrarily perturbing an eligibility threshold can change the feasible support discretely.

9 Conclusion

A hard execution-memory budget need not be replaced by an enlarged union in order to obtain a constructive guarantee. The target-crossing decomposition optimizes every alphabet budget exactly once branch targets are given. It yields global polynomial catalog design at saturation and, through a feasible weighted target net, arbitrary additive accuracy at the original budget when the number of branches is fixed. At larger branch counts, local distortion certificates and charge-aware support compression provide useful feasible policies and explicitly measured gaps without claiming that conditional optimality solves target selection.

The results retain the renewal model’s commitments: the interface carries a fixed finite alphabet, intermediate service precedes the terminal draw, and participation and realization restrictions remain separate. An execution symbol represents one enabled terminal command, while its writable bit requirement is the logarithm of the alphabet size. Opening charges model certification or maintenance of those commands; their level must be supplied or estimated for a particular institution. Zero charges describe an available interface whose extra enabled commands are treated as costless, not a factual assertion that operational memory is free. The same mathematics applies to an ordered service gateway with branch demand caps, preset command levels, an advance-service cost, and per-realization safety limits, but this is a second interpretation of the abstract model, not independent empirical validation.

The evidence supports exact conditional computation and the stated approximation and repair guarantees. The charged target-portfolio counterexample shows that a good feasible policy can still require a different target profile. A tight universal union expansion factor and scalable global approximation with an unrestricted number of branches are not established here. They are not prerequisites for the original-budget saturation theorem, the fixed-branch additive scheme, or the local certificates proved in this article.

Code and Data Availability

The anonymous reproducibility package contains the scientific source, all declared synthetic inputs and seeds, exact rational policies, independent mixed-integer formulations, raw solver outputs, and scripts that generate every new table. The root readers identify this revision; the revision directory contains its response, preservation map, and build validation. All prior mathematical sources and the complete predecessor main paper and electronic companion remain byte-preserved. Earlier executed numerical studies are archival evidence, not newly executed results. The package makes no claim to customer data, field calibration, or a deployed operational system.

A Exact Continuous Quadratic Design

Fix $s \leq m$ and temporarily allow $0 \leq c_1 \leq \dots \leq c_s \leq 1$. For each branch select one of $2s - 1$ modes. A singleton mode S_ℓ and an adjacent-lottery mode L_ℓ have the following closed descriptions:

$$S_\ell : \quad c_\ell \leq t_j \leq b_j, \quad g_j^S = rc_\ell - \frac{q}{2}c_\ell^2 - \frac{\gamma_j}{2}(t_j - c_\ell)^2; \quad (27)$$

$$L_\ell : \quad c_\ell \leq t_j \leq c_{\ell+1}, \quad t_j \leq b_j, \quad c_{\ell+1} \leq b_j + \delta, \quad g_j^L = rt_j - \frac{q}{2}[(c_\ell + c_{\ell+1})t_j - c_\ell c_{\ell+1}]. \quad (28)$$

For S_ℓ , use $y_j = t_j - c_\ell$ and terminal level c_ℓ . For L_ℓ , use $y_j = 0$ and the adjacent unbiased lottery. At $c_\ell = c_{\ell+1}$ the constraints force $t_j = c_\ell$, and the formula is the singleton reward. Therefore collisions are not discarded as singular probability formulas. The quadratic interpolation identity eliminates the ratio in the lottery probabilities before codewords become optimization variables.

Each joint mode assignment, the ordered-book inequalities, and $\sum_j \pi_j t_j = B$ define a compact polytope. If $\rho(c) = \alpha c^2 + \beta c + \eta$ is nonnegative on $[0, 1]$, its objective

$$Q(c, t) = \sum_j \pi_j g_j(c, t) - \sum_{\ell=1}^s \rho(c_\ell) \quad (29)$$

is a quadratic polynomial, not necessarily concave. Nonnegative charges, rather than signs of their coefficients, are the assumption. For example, a decreasing nonnegative certification charge is allowed.

Lemma A.1 (Finite stationary-face search). *A quadratic function on a nonempty compact polytope attains its maximum at a point that is either a vertex or the unique stationary point on the affine hull of some face with negative-definite restricted Hessian. With rational input data, some maximizing point and its value are rational and can be found by finitely many rational linear solves and feasibility tests.*

Proof. Among all faces whose relative interiors contain a global maximizer, choose one of least dimension and a maximizing point x in its relative interior. The derivative along the face's tangent space vanishes and the restricted Hessian is negative semidefinite. If this restriction has a nonzero null direction z , the quadratic is constant along $x + az$ in the affine hull: both its linear and quadratic terms in a vanish. Compactness makes this line reach a proper face while remaining feasible, contradicting the choice of dimension. The restriction is therefore negative definite unless the tangent space is zero dimensional.

Enumerate independent subsets of the defining inequalities, together with the equality constraints, and solve the stationarity equations on each associated affine space. On the face just identified the bordered stationarity matrix is nonsingular; at a vertex, the tangent space is empty and the same conclusion holds. Test feasibility in every original inequality and retain the largest objective among all feasible candidates. Each retained candidate is feasible, and at least one is a global maximizer. Rational elimination gives the rationality claim. Singular bordered systems may be skipped because the argument guarantees a nonsingular maximizing face of smaller dimension; it does not assume nondegeneracy of the original polytope. $\square$

Theorem A.2 (Exact joint continuous design). *For rational quadratic primitives, rational caps and probabilities, rational $B \in [0, \bar{b}]$ and $\delta \geq 0$, and a rational quadratic charge nonnegative on $[0, 1]$, the maximum*

of (29) over the closed cells (27)–(28), for every $1 \leq s \leq m$, equals $\mathcal{V}_m^\delta(B; \rho)$. The optimum is attained by a rational book, rational targets, and rational lottery probabilities. Deleting the lottery modes gives the exact deterministic frontier at every promise, with the same charges.

An exact algorithm enumerates at most $(2s - 1)^k$ cells at size s . After eliminating one target, each cell has dimension at most $s + k - 1$ and at most $s + 1 + 4k$ inequalities. Enumerating its independent active sets and applying Lemma A.1 gives a finite global algorithm with at most

$$\sum_{s=1}^m (2s - 1)^k 2^{s+1+4k} \quad (30)$$

linear systems, each of polynomial dimension and rational encoding length in the input size. The sharper rank-limited count replaces 2^{s+1+4k} by $\sum_{r=0}^{\min\{s+k-1, s+1+4k\}} \binom{s+1+4k}{r}$. A lottery branch may impose four separate inequalities: its two chord bounds, its target cap, and its realization eligibility bound. Neither of the latter two is redundant at a general positive tolerance. This is an exponential reference algorithm, not a polynomial-time claim or an extension of the saturated Monge complexity.

Proof. Theorem 2.1 turns every optimal book and target allocation into singleton or adjacent-lottery branch modes represented in these cells. Conversely every cell point implements a feasible policy by the displayed reconstruction; all constraints, including the exact root promise, hold. If ordered slots collide or some slots are unused, deleting redundant slots cannot increase a nonnegative charge. The resulting smaller book and its canonical branch rules appear among the enumerated sizes. Hence overcounting charges at duplicate slots cannot change the maximum across all sizes. These two directions prove equality with the original optimization problem, not merely a relaxation or a fixed-book upper bound.

There are finitely many compact cells and each has a continuous quadratic objective. Lemma A.1 supplies a rational maximizing candidate on each nonempty cell. Adjacent distinct rational levels reconstruct rational probabilities; coincident levels reconstruct a singleton. The mode count, inequality count, and all-subsets upper bound give (30). Standard determinant bounds for rational linear systems give polynomial encoding length for each candidate. Singleton-only modes exactly represent deterministic terminal assignments and intermediate tiers, proving the deterministic statement. $\square$

The reduction solves the unrestricted continuous outer problem at an interior promise, rather than relating fixed-book allocation as global design. Its contract-specific content is the eligibility-prefix reduction and the closed quadratic elimination of lottery probabilities with no loss at collisions. Rational quadratic optimization and active-set reasoning are classical; polynomial-size witnesses are not a new general complexity principle (Del Pia et al., 2017). The fast saturated Monge theorem remains valuable precisely because the general finite search is exponential.

A.1 Pathwise comparison, repeated caps, and small charges

Theorem A.2 computes $V_m^E(B)$, $V_m^P(B)$, and $V_m^D(B)$ for every rational promise, using respectively all lottery modes at $\delta = 1$, all modes at $\delta = 0$, and singleton modes. No claim of pathwise–deterministic equality away from saturation is needed for this characterization. The complete predecessor companion retains the

earlier exact institutional comparison path; the current study reports the separately executed budgeted-design comparisons. The zero-memory-loss interval $B \leq \min b_j$ in Theorem EC.4.1 remains important: a positive institution gap is not expected throughout the feasible promise range.

Proposition A.3 (Pathwise randomization can matter in the interior). *For one branch with $b = 9/10$, $B = 9/20$, $r = 2$, $q = \gamma = 1$, two available symbols, and the nonnegative charge $\rho(c) = 3c(1 - c)$,*

$$\mathcal{V}_2^0(B; \rho) = \frac{171}{400} > \mathcal{V}_2^D(B; \rho) = \frac{9}{160}.$$

The pathwise optimum uses book $(0, 9/10)$, equal draw probabilities, and no intermediate tier. The deterministic optimum uses the single level $9/20$.

Proof. A deterministic policy uses only one level $0 \leq c \leq B$; deleting unused levels weakly lowers its charge. Its net payoff is $-81/800 - (11/20)c + 2c^2$, whose maximum over $[0, B]$ is $9/160$ at B . Every nonsingleton canonical pathwise mode has $0 \leq u \leq B \leq v \leq b$, and net payoff $2B - (u+v)B/2 + uv/2 - 3u(1-u) - 3v(1-v)$. This is convex in (u, v) , so its maximum on that rectangle is at a corner. Checking its four corners, together with the singleton case, gives $(u, v) = (0, 9/10)$ and value $171/400$. Both realizations meet the hard cap and their mean is exactly B . $\square$

This example uses a specified nonnegative, nonmonotone level charge; it is not a counterexample asserted for the uncharged problem. It shows why the saturated pathwise collapse cannot simply be imported into the unified interior design model.

Proposition A.4 (Homogeneous repeated-cap aggregation). *For the randomized institutions in (1), branches with identical caps and identical intermediate-cost functions can be aggregated by adding their probabilities, at every B , δ , and codeword charge. This holds for each fixed book and after global book optimization.*

Proof. Such branches have identical eligible prefixes and the same concave response (2). Replacing their targets by their probability-weighted mean preserves the root promise and does not lower total reward by Jensen's inequality. Replicating an aggregate target gives the converse inequality. Charges depend only on the common book. This argument does not aggregate heterogeneous costs or establish a corresponding deterministic restriction. $\square$

The strict off-cap conclusion is also stable under genuinely selected-level charges. In Proposition B.3, the uncharged two-level book $(1/4, 5/8)$ beats every book of at most two cap levels by $1/1280$. If $0 \leq \rho(c) \leq \eta$ on $[0, 1]$ and $\eta < 1/2560$, evaluating this same off-cap book incurs less than that gap, while a cap-only book cannot improve its uncharged payoff after a nonnegative charge. Thus no charged optimum is cap-only. This statement preserves the off-cap property, not necessarily the numerical location $5/8$.

B The Global Continuous Lottery Codebook

The fixed-codebook formula in Theorem EC.3.1 leaves a continuous design problem: both the terminal service levels and their assignment probabilities must be chosen. We now solve that problem globally for

quadratic terminal rewards and heterogeneous quadratic intermediate costs. The solution also determines when imposing participation after, rather than before, the private draw eliminates the value of randomization. These are restricted-alphabet design results, distinct from the exact-implementation alphabet in Theorem EC.4.1.

Retain the three-date renewal architecture of Theorem EC.2.1, including the saturated root promise $\bar{b} = \sum_{j=1}^k \pi_j b_j$, strictly ordered caps $0 < b_1 < \dots < b_k < 1$, and positive probabilities summing to one. Throughout this section,

$$f(c) = rc - \frac{q}{2}c^2, \quad h_j(y) = \frac{\gamma_j}{2}y^2, \quad r \geq q > 0, \quad \gamma_j \geq 0. \quad (31)$$

The parameters and probabilities can differ across branches as indicated. Terminal reward is strictly increasing on the feasible cap range, including when $r = q$. All private information retained until the common terminal vertex must be encoded in the charged symbol. The decoder and public observation convention are unchanged.

B.1 Cap anchors and one continuous terminal level

For $u \leq b \leq v$, quadratic concavity gives the interpolation loss

$$f(b) - \frac{v-b}{v-u}f(u) - \frac{b-u}{v-u}f(v) = \frac{q}{2}(b-u)(v-b), \quad (32)$$

with the endpoint value understood when levels coincide. Thus an interior codeword enters the loss affinely until it crosses a branch cap. The highest codeword is different: reducing it below a branch cap creates an intermediate payment and its associated cost.

Lemma B.1 (Cap-anchored representation). *For every alphabet budget $m \geq 1$, an optimum of (EC.5) exists with the following representation. Its lowest codeword is b_1 ; every codeword except the highest belongs to $\{b_1, \dots, b_k\}$; and the highest belongs to $[b_1, b_k]$. A one-symbol optimum is $\{b_1\}$. The highest codeword need not be a branch cap.*

The proof is in Electronic Companion Section EC.5.4.

The lemma does not discretize the highest level. A cap-only restriction can be strictly suboptimal, as Proposition B.3 shows. It also does not assert that every optimal codebook is anchored; it supplies an optimum with this useful form.

B.2 A finite exact algorithm

For adjacent anchored levels $b_i < b_j$, define

$$E(i, j) = \frac{q}{2} \sum_{h=i}^j \pi_h (b_h - b_i)(b_j - b_h). \quad (33)$$

Let $P(s, j)$ be the minimum interpolation loss on caps through b_j using exactly s anchored levels, starting at b_1 and ending at b_j . Set $P(1, 1) = 0$, $P(1, j) = +\infty$ for $j > 1$, and use the ordered recurrence

$$P(s, j) = \min_{i < j} \{P(s-1, i) + E(i, j)\}, \quad 2 \leq s \leq j. \quad (34)$$

This is a standard shortest-path or segmentation recurrence. The substantive reduction is Lemma B.1, which makes it applicable to a jointly optimized continuous lottery codebook.

Suppose that b_i is the last anchored level and z is the final level. For $\ell \geq i$ and $b_\ell \leq z \leq b_{\ell+1}$, put $b_{k+1} = b_k$ and define

$$\begin{aligned} \Psi_{i\ell}(z) = & \frac{q}{2} \sum_{h=i+1}^{\ell} \pi_h (b_h - b_i)(z - b_h) \\ & + \sum_{h=\ell+1}^k \pi_h \left\{ f(b_h) - f(z) + \frac{\gamma_h}{2} (b_h - z)^2 \right\}. \end{aligned} \quad (35)$$

The first sum is the lottery loss below z ; the second is the terminal-reward and intermediate-payment loss above it. Empty sums are zero. For $\ell < k$ the function is strictly convex in z and has the constrained minimizer

$$z_{i\ell} = \left[\frac{r \sum_{h>\ell} \pi_h + \sum_{h>\ell} \pi_h \gamma_h b_h - \frac{q}{2} \sum_{h=i+1}^{\ell} \pi_h (b_h - b_i)}{q \sum_{h>\ell} \pi_h + \sum_{h>\ell} \pi_h \gamma_h} \right]_{[b_\ell, b_{\ell+1}]}. \quad (36)$$

For $\ell = k$, take $z_{ik} = b_k$. Define the terminal cost

$$T_i = \min_{i \leq \ell \leq k} \Psi_{i\ell}(z_{i\ell}). \quad (37)$$

Theorem B.2 (Global frontier and exact complexity). *Under (31), the optimum with at most m symbols is*

$$D_m^{\text{ran}} = \min \left\{ D_1, \min_{\substack{1 \leq s \leq \min(m-1, k) \\ 1 \leq j \leq k}} \{P(s, j) + T_j\} \right\}, \quad (38)$$

where the inner minimum is empty for $m = 1$. Backtracking produces a globally optimal codebook and the adjacent lotteries implementing it. For ordered inputs and $1 \leq m \leq k$, all frontiers through budget m can be computed in $O(mk^2)$ exact rational operations and $O(mk + k)$ stored rational numbers and indices, including backpointers. No codeword grid is required.

For rational input of total binary length L , a reduced-rational implementation has polynomial Turing complexity. A conservative bound is $O(mk^2 S^3)$ bit operations with $S = O(k[L + \log(k+2)])$. This includes exact cost comparisons, clamping, and fraction reduction. For $m \geq k$ the optimal loss is zero; listing additional budget entries requires only their output cost. These guarantees concern the stated algorithm, not an arbitrary nonlinear optimizer for (EC.5).

The proof is in Electronic Companion Section EC.5.5.

B.3 A continuous optimal highest level

Proposition B.3 (An off-cap terminal level is optimal). *Let $(b_1, b_2, b_3) = (1/4, 1/2, 3/4)$, $(\pi_1, \pi_2, \pi_3) = (7/20, 3/5, 1/20)$, $r = 2$, $q = 1$, and $\gamma_j = 1$. The globally optimal two-symbol expected-participation codebook is $(1/4, 5/8)$, and*

$$D_2^{\text{ran}} = \frac{23}{1280} < \frac{3}{160} = \min_{\substack{|c| \leq 2 \\ c \subseteq \{b_1, b_2, b_3\}}} D(c) = D_2, \quad (39)$$

where the cap-only minimum is over feasible codebooks and $D(c)$ denotes their expected-participation loss. Thus even optimizing all cap-only lotteries misses the unrestricted optimum.

The proof is in Electronic Companion Section EC.5.6.

B.4 Participation after the private draw

The preceding frontier is appropriate when a renewal review approves an expected continuation before the protocol draws its private symbol. Consider instead an agreement that requires the same cap to hold for every private realization after that review. A draw-specific intermediate tier is allowed; only the terminal public state and the charged symbol reach the terminal decoder. Write $D_m^{\text{pw,ran}}$ for the best randomized loss under these pathwise constraints and the same saturated root promise.

Theorem B.4 (Pathwise randomization has no frontier advantage). *For the general renewal family of Theorem EC.2.1, with strictly increasing concave f and convex nondecreasing h_j , and the observation architecture just specified,*

$$D_m^{\text{pw,ran}} = D_m \quad \text{for every } m \geq 1. \quad (40)$$

Consequently (38) and the deterministic frontier give an exact comparison of the two participation institutions under (31). Their exact-optimum alphabet thresholds are both k when strict concavity makes the k distinct full-information actions unique.

The proof is in Electronic Companion Section EC.5.7.

This institutional result is not an assertion that randomization is generally ineffective in dynamic contracts. It uses the saturated root promise, an exogenous recombination, and the charged-information interface. Under expected participation, Theorem B.2 chooses a less expensive lottery protocol; under pathwise participation, the same service model calls for the deterministic frontier. In either institution an operator can price a symbol budget, add its implementation charge to the computed loss, and optimize the budget without confusing read-only program size with retained private information.

C Monge Costs and Classical Linear Matrix Search

The finite dynamic program still optimizes a continuous highest service level. We first prove the contractual cost inequalities and then verify the classical easy-direction completion with the exact order used by the code. A search routine cannot establish the economic reduction that supplies its entries.

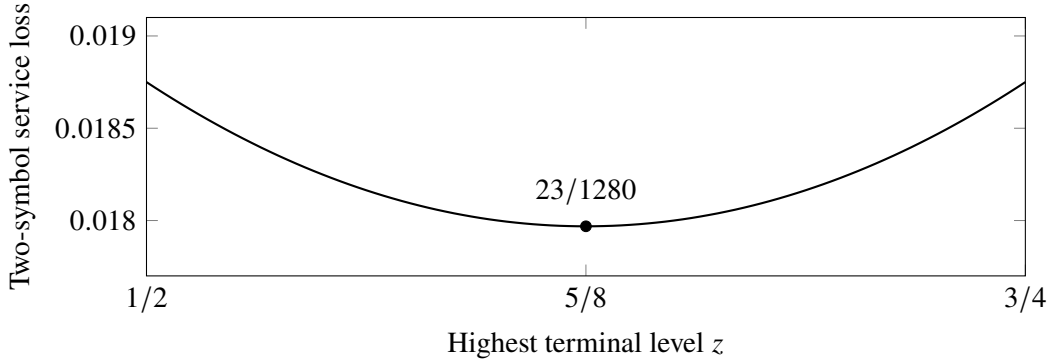


Figure 1: The analytically optimized highest level in the heterogeneous three-branch example. The lower level is $1/4$. The displayed function is $z^2/20 - z/16 + 3/80$ on $[1/2, 3/4]$; its interior minimum is $z = 5/8$, not a branch cap.

C.1 The inequality survives continuous terminal minimization

Write $I_\ell = [b_\ell, b_{\ell+1}]$, including $I_k = \{b_k\}$, and $Q(i, \ell) = \min_{z \in I_\ell} \Psi_{i\ell}(z)$ for $i \leq \ell$. A partial array is Monge here when

$$A(r, c) + A(s, d) \leq A(r, d) + A(s, c), \quad r < s, c < d, \quad (41)$$

whenever all four entries are defined. This fixes the inequality convention before any orientation is named.

Lemma C.1 (Contractual finite-cost inequalities). *The anchored edge costs E and deterministic cell costs C satisfy $A(a, c) + A(b, d) \leq A(a, d) + A(b, c)$ for $a \leq b \leq c \leq d$. The continuously minimized terminal array Q is Monge on its feasible domain. These statements permit arbitrary positive probabilities and heterogeneous nonnegative intermediate curvatures.*

The proof is in Electronic Companion Section EC.5.8.

Lemma C.2 (Previous-layer values are column offsets). *Adding any finite column term v_c preserves the Monge inequality. For fixed finite predecessor values, the arrays $A(j, i) = P(s, i) + E(i, j)$ and $B(j, i) = D(s - 1, i) + C(i + 1, j)$ are therefore Monge wherever all four entries are feasible.*

The proof is in Electronic Companion Section EC.5.9.

C.2 Completion with the implemented strict and tie orders

Give a feasible entry the lexicographic key

$$K(r, c) = (0, A(r, c), c). \quad (42)$$

Finite ties select the smaller column, and distinct columns have distinct keys. The exact total-monotonicity condition used by the implementation is

$$K(r, d) < K(r, c) \implies K(s, d) < K(s, c), \quad r < s, c < d. \quad (43)$$

A strict preference for the right column cannot reverse in a later row. This applies to arbitrary selected rows and columns. Because keys are distinct, a decrease of row-argmin indices in any submatrix would contradict (43).

Lemma C.3 (Suffix-feasible completion). *Let row r have feasible columns $c \geq \ell_r$, with nondecreasing cutoffs ℓ_r . Suppose the finite entries are Monge. Complete every $c < \ell_r$ with $K(r, c) = (1, 0, -c)$. Then (43) holds. A completely infeasible row of a selected submatrix selects its largest selected column.*

The proof is in Electronic Companion Section EC.5.10.

Lemma C.4 (Prefix-feasible completion). *Let row r have feasible columns $c \leq u_r$, with nondecreasing cutoffs u_r . Suppose the finite entries are Monge. Complete every $c > u_r$ with $K(r, c) = (1, 0, c)$. Then (43) holds. A completely infeasible selected row selects its smallest selected column.*

The proof is in Electronic Companion Section EC.5.11.

These lemmas specialize the known favorable completion setting in Section 1.1; they are not general partial-matrix search theorems. A helper accepting two bounds does not prove total monotonicity for arbitrary interval rows. For example, two zero-cost rows feasible only at columns 2 and 1 have a vacuously Monge finite part but decreasing minima. Uniform infinite padding with ordinary left ties also fails on the required suffix domain: rows with cutoffs 2, 3 restricted to columns 1, 2 select 2, 1. The opposite internal padding orders above are deliberate.

C.3 Reachability and three search interfaces

Lemma C.5 (Contiguous reachable predecessor sets). *For the anchored program, $P(1, j)$ is finite exactly at $j = 1$, and for $s \geq 2$, $P(s, j)$ is finite exactly when $s \leq j \leq k$. For the deterministic program, $D(0, j)$ is finite only at $j = 0$, and $D(s, j)$ is finite exactly when $s \leq j \leq k$ for $s \geq 1$. Each globally reachable predecessor set that can extend to a further endpoint is a contiguous interval or the initial singleton.*

The proof is in Electronic Companion Section EC.5.12.

The following pseudocode uses one-based cap indices and zero for the empty deterministic prefix. “Search” means ordinary full-matrix SMAWK on the supplied key. All prior-layer values are known before a search.

Terminal interface. Rows and columns are $1, \dots, k$. At (i, ℓ) return $(1, 0, -\ell)$ if $\ell < i$; otherwise minimize (35) by (36) and return $(0, Q(i, \ell), \ell)$. Search all rows. Reevaluate each winning cell once to store T_i and its continuous optimizer.

Anchored-prefix interface. To form layer $s+1$, use rows $j = s+1, \dots, k$. Columns are $\{1\}$ for $s = 1$, and $s, \dots, k-1$ otherwise. At (j, i) return $(1, 0, i)$ if $i \geq j$, and $(0, P(s, i) + E(i, j), i)$ otherwise. Search and store values and predecessors. Scan the current reachable endpoints with T_i to update the at-most-budget incumbent; backtrack only its winning codebook.

Deterministic-prefix interface. To form layer s , use rows $j = s, \dots, k$. Columns are $\{0\}$ for $s = 1$, and $s-1, \dots, k-1$ otherwise. Return $(1, 0, i)$ if $i \geq j$, and $(0, D(s-1, i) + C(i+1, j), i)$ otherwise. Search, store predecessors, and backtrack the solution ending at k .

Lemma C.5 justifies every discarded column. The terminal interface has shrinking feasible suffixes; the program interfaces have expanding feasible prefixes. The completion lemmas include infeasible rows in submatrices created internally by SMAWK, although every original searched row has a feasible entry. Padded comparisons do not evaluate an undefined economic cost.

Theorem C.6 (Contractual corollary of classical matrix search). *For ordered rational inputs satisfying (31) and $1 \leq m \leq k$, both expected-participation and deterministic/pathwise frontiers through budget m , including an optimal codebook for every budget, require $O(mk)$ rational operations and comparisons and $O(mk + k)$ stored rational numbers and indices. The highest randomized level remains continuous. With total input length L and $S = O(k[L + \log(k + 2)])$, a conservative Turing bound is $O(mkS^3)$. Sorting unordered distinct caps adds $O(k \log k)$ exact comparisons.*

The proof is in Electronic Companion Section EC.5.13.

The retained divide-and-conquer implementation is a valid $O(mk \log(k + 1))$ alternative using ordinary row-minimum monotonicity. It is an implementation comparator rather than another headline theorem. $O(k)$ exact-rational oracle/comparison work per layer does not imply uniform wall-time superiority for a particular Python implementation.

D Catalog Approximation Without Promise or Risk Slack

Finite catalogs have two different interpretations. An externally certified menu is a restriction of the feasible technology, for which an exact catalog optimum is the relevant answer. Alternatively the operator may choose a numerical catalog to approximate continuous design. The latter interpretation needs a quantitative comparison to the unrestricted optimum; exactness on the catalog alone does not supply it.

Let $\mathcal{A} \subset [0, 1]$ contain zero and one, and let h be its largest consecutive spacing. Assume f is increasing, concave, and L_f -Lipschitz, and each h_j is convex, nondecreasing, and H_j -Lipschitz on $[0, b_j]$. Let the nonnegative charge ρ be L_ρ -Lipschitz on $[0, 1]$. Define

$$K_m = L_f + \sum_j \pi_j H_j + m L_\rho. \quad (44)$$

For the quadratic model, one can take $L_f = r$ and $H_j = \gamma_j b_j$. Neither the root promise nor the risk tolerance is discretized.

Theorem D.1 (Uniform exactly feasible catalog certificate). *For every $B \in [0, \bar{b}]$, every $\delta \geq 0$, and every $m \geq 1$,*

$$0 \leq \mathcal{V}_m^\delta(B; \rho) - \mathcal{V}_{m, \mathcal{A}}^\delta(B; \rho) \leq K_m h. \quad (45)$$

More strongly, every canonical continuous-book policy has a catalog policy with no more symbols that preserves each branch target exactly, satisfies the same realization ceiling exactly, and loses at most $K_m h$ in net payoff. The conclusion includes the pathwise endpoint $\delta = 0$ and requires no strict feasibility margin.

Proof. Round each original level c downward to $a(c) = \max\{a \in \mathcal{A} : a \leq c\}$ and merge collisions. Fix a branch target t and let d be its old eligible top. Every old eligible level remains eligible after

rounding. In particular the new eligible top d' satisfies $d' \geq a(d) \geq d - h$. The old canonical terminal mean is $\mu = \min(t, d)$; coupling its draws to their rounded levels gives a feasible-support lottery of mean $\tilde{\mu} \in [\mu - h, \mu]$ and expected terminal reward at least the old reward minus $L_f h$.

Do not retain those coupled probabilities. Recompute the canonical lottery on the rounded eligible prefix at the *same* target t . Its terminal mean $\mu' = \min(t, d')$ is at least $\tilde{\mu}$: every rounded old draw is at most d' and its mean is at most t . The new interpolant is increasing, and at mean $\tilde{\mu}$ it dominates the coupled lottery reward. Therefore its reward at μ' also dominates that reward. Moreover $\mu' \geq \mu - h$, so its intermediate tier $t - \mu'$ exceeds the old tier by at most h . The branch operating-payoff loss is at most $(L_f + H_j)h$. Theorem 2.1 ensures exact mean and risk feasibility for this reoptimized lottery, including when the risk bound is binding. All t_j are unchanged, so the root promise remains exactly B .

For each original level, $\rho(a(c)) \leq \rho(c) + L_\rho h$. After merging, the total charge is no larger than the sum of these rounded-slot charges because charges are nonnegative. At most m slots were present. Summing the branch bounds and charge bound gives $K_m h$. Apply the construction to an optimizer, or to a sequence approaching the supremum, and use inclusion of catalog policies in continuous policies for the other inequality. $\square$

One-sided rounding is necessary but not sufficient without the probability repair. For example, let $b = t = 1/2$, $\delta = 1/8$, and old support $(1/4, 5/8)$ with upper probability $2/3$. Rounding the lower level to zero while retaining the probabilities reduces the mean to $5/12$. Compensating by $y = 1/12$ produces a maximum obligation $17/24 > 5/8$, violating the risk limit. Instead, the new canonical lottery on $(0, 5/8)$ uses upper probability $4/5$ and $y = 0$. It preserves the target and the ceiling exactly. This is the distinction between unconstrained quantizer rounding and the contractual transport in Theorem D.1.

Corollary D.2 (Certified approximation for a fixed alphabet budget). *For rational quadratic operating primitives and a nonnegative charge whose rational catalog evaluations and Lipschitz bound are supplied, enumerate every feasible catalog book of size at most m and solve its allocation by (4). With $N = |\mathcal{A}|$, this gives an exact catalog optimum in*

$$O\left(\left(\sum_{s=1}^{\min(m, N)} \binom{N}{s}\right) (km + k \log(k+1))\right) \quad (46)$$

rational arithmetic/evaluation operations. A uniform mesh with denominator $\lceil K_m/\varepsilon \rceil$ (or one when that ceiling is zero) gives additive error at most ε . The algorithm is polynomial in k , K_m/ε , and the bit complexity of the sampled data for fixed m ; it is not claimed to be polynomial in a variable m , nor is this a multiplicative approximation ratio.

Proof. Increasing subsets exhaust the allowed books and the supporting-price algorithm optimizes every inner problem exactly. Take the largest charge-adjusted value and apply Theorem D.1. The mesh includes zero, so every feasible promise is implementable even when it is smaller than the first positive grid point. The bound counts evaluation work; a black-box charge oracle is not assumed to have unit bit complexity unless that is explicitly part of the input model. $\square$

The guarantee is uniform in the two contractual constraints. For any institution pair, subtracting their certified value intervals gives a certified interval for the institutional gap. This permits a global accuracy statement away from saturation without presuming that original cap anchors or Monge order survive. When $B = \bar{b}$, the faster charged path recurrence in Section EC.9 replaces subset enumeration. When the saturated expected problem is uncharged, the continuous Monge algorithm avoids a mesh entirely. Thus the specialized algorithms are faster members of one feasible-policy hierarchy, while (45) connects numerical catalogs to continuous risk-and-charge design.

D.1 The uniform-mesh rate is sharp

The linear bound in Theorem D.1 reflects exact feasibility, not only an insensitive Lipschitz estimate. It cannot be uniformly replaced by a higher-order rate for the declared uniform catalogs.

Proposition D.3 (A fixed-input lower bound). *Take one branch and one symbol, $b = B = 1/2$, $f(c) = c - c^2/2$, $h(y) = y^2/2$, and zero charges. For every risk tolerance and every odd integer $N \geq 3$, the uniform catalog with spacing $h = 1/N$ has loss*

$$\mathcal{V}_1^\delta(B; 0) - \mathcal{V}_{1, \mathcal{A}_N}^\delta(B; 0) = \frac{h}{4} + \frac{h^2}{4}.$$

Thus the worst-case $O(h)$ rate is order-sharp, even for a single fixed input with no opening charges and no randomization.

Proof. With one level, $c \leq B$ and $y = B - c$. Payoff $f(c) - (B - c)^2/2$ is strictly increasing on $[0, B]$. Its continuous maximizer is B , with value $3/8$. For odd N , the largest feasible catalog point is $B - h/2$. Substitution gives the displayed loss. All realizations equal total service B , so risk tolerance plays no role. The sequence of odd N uses the same primitives throughout. $\square$

This is a lower bound for uniform catalogs, not for all adaptive discretizations. Adding the cap $1/2$ removes this particular error. It also does not rule out faster convergence on other fixed instances. The search interval in (23) is therefore useful even when its optimization gap has already vanished: refinement addresses a genuinely different source of error.

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Table 1: Original-budget comparison across six seeds

Charges	Cases	Core exact	Safe exact	Core wins	Greedy wins	Max core loss
None	18	18	18	2	0	0.000000
Nonuniform	18	17	18	8	1	0.028056

Exact means equality with the exhaustive joint catalog reference. Core and greedy wins compare their same-budget values; ties are omitted. The safeguarded policy keeps the better feasible candidate. The charged exception remains in the data. These counts are observations, not global guarantees for a finite target portfolio.

Table 2: Completed same-budget target-net comparisons

Branches	Tolerance	Cases	Profiles	Maximum actual loss
2	1/4	4	28	0.000000
2	1/8	4	50	0.000000
2	1/16	4	96	0.000000
3	1/2	4	47	0.049805
3	1/4	4	164	0.034645

All completed nets meet the requested additive tolerance with the original two-symbol budget and nonuniform charges. Counts aggregate four seeds. An additional interrupted run withholds its global accuracy guarantee.

Table 3: Conditional frontier scaling with heterogeneous realization ceilings

Seed	Branches	Catalog	Budget	Charges	Used	Seconds	Gap
104729	32	17	4	No	4	0.036	0.019173
104729	32	17	4	Yes	4	0.036	0.037009
104729	128	33	8	No	8	0.486	0.004410
104729	128	33	8	Yes	5	0.484	0.041534
104729	256	65	8	No	8	3.257	0.004923
104729	256	65	8	Yes	8	3.278	0.025671
205019	32	17	4	No	4	0.038	0.010627
205019	32	17	4	Yes	3	0.038	0.040935
205019	128	33	8	No	8	0.474	0.003661
205019	128	33	8	Yes	5	0.478	0.042000
205019	256	65	8	No	8	3.303	0.004099
205019	256	65	8	Yes	7	3.316	0.023976

Seconds are medians of three repeated executions on the same input. Gap is the valid same-budget Jensen upper bound minus the ideal-target conditional policy value, not an observed error to an unknown global optimum. Full timing arrays, stored-entry bit lengths and cumulative process resident-memory records are archived.

Table 4: Adaptive candidates at an unchanged installed-symbol budget

Case	Candidate pool	Candidates	Symbols	Exact-reference loss
interior-ideal	uniform	9	2	0.044030
interior-ideal	uniform	65	2	0.002991
interior-ideal	target-adaptive	4	2	0.000000
off-cap-top	uniform	9	2	0.000000
off-cap-top	Adaptive	5	2	0.000781
off-cap-top	Adaptive	6	2	0.000000
Sat. 104729	uniform	9	3	0.006507
Sat. 104729	Adaptive	9	3	0.000000
Sat. 205019	uniform	9	3	0.001824
Sat. 205019	Adaptive	9	3	0.000000

Every row has an exact continuous reference in the stated Jensen-attaining or saturated expected-participation regime. Candidate-pool size is not installed memory. Adaptive midpoint selection is a heuristic; catalog optimization at the saturated target is exact. The complete 22-row path is in the companion and JSON.